

1.0 EXECUTIVE SUMMARY

Detection of malware is a way of decision making in which the malware should be detected and identified by the end of the procedure (M. & Sethuraman, 2023). This paper shows a clear and deep understanding of the malware detection techniques. In addition to this, the methods of finding and analyzing the malware are shown in three different ways including Hybrid-based detection, signature-based detection and behavior-based detection. This also offers an overview of the existing problems related with methods for detecting malware. This paper examines the framework of important techniques needed for the identification of malware. The detection techniques of malware have been compared with important functionalities including the advantages and disadvantages of each approach.

1.1 MALWARE DETECTION

Malware analysis basically examines the file, URL or hash in order to understand its behaviors and its specific function (Baker, 2023). The sample identification of malwares can be done by the malware detection techniques. A malware is any software or program which can be programmed by cybercriminals that is specially created to harm, disrupt, or gain unauthorized access to computer systems or devices (What is malware? 2023). If a malware is not identified there can be a huge damage to the systems.

2.0 SIGNATURE BASED MALWARE DETECTION METHOD

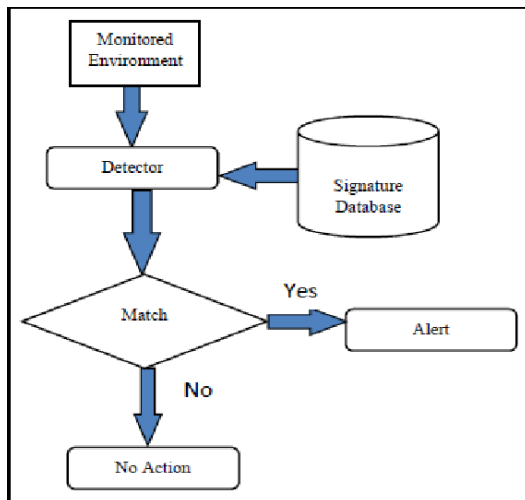


Figure 1.0

The malware has unique characteristics and also has a unique digital signature. Signature-based malware detection method detects malware by comparing files to the unique signatures or patterns related to a database containing the signatures of previously discovered malware (Chakravarty et al., 2019). If the signatures are matching, the file is labeled as malware. This important procedure is called as the static analysis (Souri & Hosseini, 2018). The static analysis gives needed information of the malicious URLs, related IP addresses, file type and history of the origin of malware (Madan et al., 2022). Which can be used to find when and why the malware was created. Malware without a signature registered in the database will not be detected by signature-based detection. Additionally, if a machine is attacked by another form of the same virus, this method loses its effectiveness. As the outcome, it is unable to detect malwares if it's in various forms. On the other hand, this detection technique is great for the known threats.

This table is written according to (Souri & Hosseini, 2018).

Advantages	Disadvantages
Simple to proceed	The polymorphic malwares are unable to be detected.
Can be identified quickly	Copies data in the large database.
Widely available	
Detects data deeply related to malware.	

Table 1.0

3.0 BEHAVIOUR BASED MALWARE DETECTION METHOD

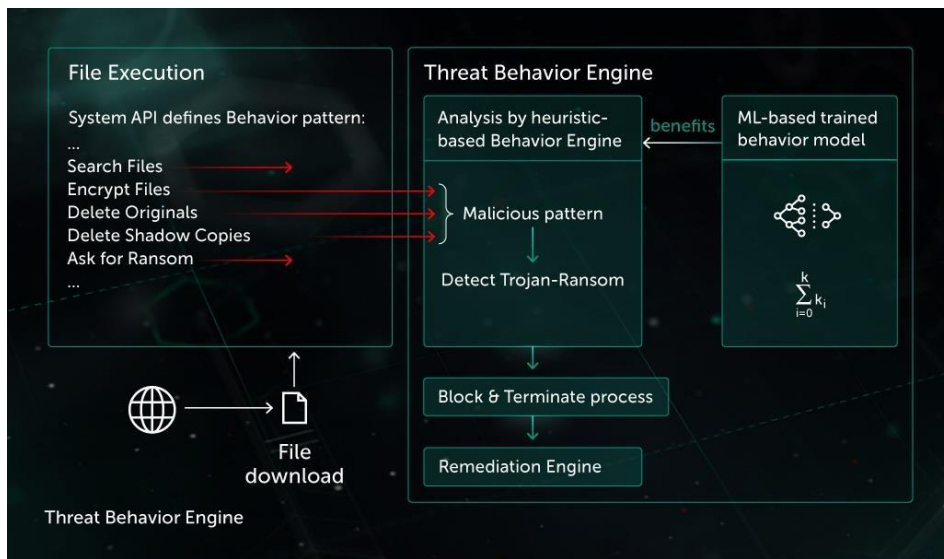


Figure 1.1

As the signature-based method is a static analysis approach the behaviour-based detection is a dynamic analysis approach (Devesa et al., 2010). This allows the malware to be run on a specific environment in which the functions are observed and analyzed. This approach is done on a special and isolated place because if it's done on the main system, it might bring a great damage to the existing systems and to the connected networks (Madan et al., 2022). This analysis is basically done in a virtual machine or a stimulator (Sihwail et al., 2018). In contrast to the signature-based detection technique this technique is very helpful for detecting and identifying unknown malwares. Antivirus scanners need to conduct their own automatic analysis in order to detect the malware before it has been thoroughly examined. Somehow some researches need to be behavior-based since the malware will always behave badly, regardless of its disguise (Elhad et al., 2012). So, the behavior-based malware analysis plays a huge role in modern days.

The polymorphic malware has a polymorphic code as the original code, but each mutation has a different appearance. This is made for the detection to be harder and lot of malwares use this technique (Ahmadi et al., 2013).

This table 2.0 is written according to (Souri & Hosseini, 2018).

Advantages	Disadvantages
Detecting the previously found attacks.	Complexity of storage for behavioral patterns.
Dependency detection for the flow of data.	Consumes more time.
Polymorphic malwares are identified.	

Table 2.0

4.0 HYBRID BASED MALWARE DETECTION METHOD

This technique is a combination of both signature-based and behavior-based detection techniques. So, both the advantages of static and dynamic approaches are available in this method (Sihwail et al., 2018). The overall accuracy rate and the problems in time consuming are solved in this method (Damodaran et al., 2015). Similar to the signature-based detection technique this method is also very useful to identify the known malwares and as the behavior-based detection technique this method is also very useful to identify the unknown malwares. According to (Roundy et al., 2010) research, analysis-guided instrumentation on obfuscated, packed, and self-modifying program binaries in malware are achieved by the mixture of static and dynamic analysis. This itself shows the hybrid-based detection method is definitely helpful for the upcoming generations in achieving new tasks. From the experiment done by (Sihwail et al., 2018) to detect the unknown files the hybrid analysis has provided a 98.1% of accuracy rate than compared to the other techniques of static analysis which provided 95.8% and dynamic analysis which resulted in 98.7%. In contrast, this method requires more resources and there can be also a complexity in using two techniques together.

5.0 COMPARISON OF THE 3 TECHNIQUES

SIGNATURE-BASED	BEHAVIOR-BASED	HYBRID
Simple to proceed.	Hard to proceed.	Very hard to proceed.
Need less resources.	Need more resources.	Needs the greatest number of resources.
Accuracy rate is very high.	Accuracy rate is high	Gives the greatest accuracy rate.
Files are not executed or monitored.	Files are executed and monitored.	Files are executed and monitored.
Not good in detecting unknown malware.	Very good in detecting unknown malware.	Very good in detecting unknown malware.
Less false positives.	False positives vary according to the malware type.	Can give more compared to the other two techniques.
It is not adaptable to new threats as if the signatures are not matching the malware will not be identified. So, it depends on the signatures available.	It is adaptable to new threats as it monitors the running malware.	It is very adaptable to new threats as it monitors the running malware and also uses the signatures to detect malwares.
Gives more information on known malwares.	Gives less information on known malwares compared to signature-based detection method.	Gives less information on known malwares compared to signature-based detection method.
Faster to proceed.	Takes more time.	Takes more time.
May not be very effective in future and newly upcoming malwares.	Can be effective in future and upcoming malwares	Can be effective in future and upcoming malwares
Simple to maintain.	Hard to maintain.	Hard to maintain.
Easier to understand as all outputs are from a previously detected malware.	Difficult to understand as proceeds on an isolated environment which should be monitored.	It is harder to understand compared to signature-based detection but easier to understand than behavior-based detection.

Implementation is easier.	Hard to implement it as it involves configuring systems to monitor and analyzing various aspects of behavior, such as process functions, file access, and communication in network.	Very hard to implement as it involves the combination of other two techniques.
Uses the real time files which are readily available.	Mostly used after an incident takes place. It can be also used in real time.	Mostly used in real-time but can also be used after an incident has occurred.
Needs regular updates in database as the new signatures should be added when new malwares found.	Needs regular updates on the detection algorithms to detect the malwares.	Needs regular updates on signatures and also needs updates on detection algorithms.
No need to be trained.	Need to be trained.	Need to be trained.
Works on the endpoint level.	Can work on both: endpoint level and also inside the network.	Can work on both: endpoint level and also inside the network.
Hard to examine encrypted data without decryption.	Can detect the encrypted data in malware without decrypting the data.	As it uses both the techniques it can be very effective to detect the encrypted data in malware.
Not very adaptable to the changes in system as it can even result in false positives.	Very good in adapting to the changes in system.	Very good in adapting to the changes in system.
May not detect human interactions.	Can be able to detect human interactions.	Can be able to detect human interactions.

Table 3.0

5.1 CONCLUSION

The state-of-the-art detection techniques play a great role in modern days as the threat of malwares has also increased. The three detection techniques mentioned above has its weaknesses and strengths. Signature-based detection and behavior-based detection both are unique ways of detecting malware in which both follow a different pattern which is not relevant to each other. The hybrid-based detection is a combination technique of both. The hybrid-based method can be very effective in future and doing modifications can result in great advancements.

6.0 REFERENCES

- What is malware?* (2023). From CISCO:
<https://www.cisco.com/site/us/en/products/security/what-is-malware.html#:~:text=Malware%2C%20short%20for%20malicious%20software,spyware%2C%20adware%2C%20and%20ransomware.>
- Baker, K. (2023, April 17). *What is Malware Analysis?* From CROWDSTRIKE:
<https://www.crowdstrike.com/cybersecurity-101/malware/malware-analysis/#:~:text=Static%20Analysis&text=It%20can%20be%20useful%20to,whether%20that%20file%20is%20malicious>
- Damodaran, A., Troia, F. D., Visaggio, C. A., Austin, T. H., & Stamp, M. (2015, December 29). A comparison of static, dynamic, and hybrid analysis for malware detection. *Journal of Computer Virology and Hacking Techniques*, 13(1), 1–12.
<https://doi.org/10.1007/s11416-015-0261-z>
- Sihwail, R., Omar, K., & Zainol Ariffin, K. A. (2018, September 30). A Survey on Malware Analysis Techniques: Static, Dynamic, Hybrid and Memory Analysis. *International Journal on Advanced Science, Engineering and Information Technology*, 8(4–2), 1662. <https://doi.org/10.18517/ijaseit.8.4-2.6827>
- Ahmadi, M., Sami, A., Rahimi, H., & Yadegari, B. (2013, August). Malware detection by behavioural sequential patterns. *Computer Fraud & Security*, 2013(8), 11–19.
[https://doi.org/10.1016/s1361-3723\(13\)70072-1](https://doi.org/10.1016/s1361-3723(13)70072-1)
- Aubrey-Jones, T. (2007). Behaviour Based Malware Detection. *Computer*, 3-5.
- Chakravarty, A. K., Raj, A., Paul, S., & Apoorva, S. (2019). A study of signature-based and behaviour-based malware detection approaches. *Int. J. Adv. Res. Ideas Innov. Technol.*, 5(3), 1509-1511.
- Devesa, J., Santos, I., Cantero, X., Penya, Y. K., & Bringas, P. G. (2010, June). Automatic behaviour-based analysis and classification system for malware detection. In *International Conference on Enterprise Information Systems* (Vol. 2, pp. 395-399). SCITEPRESS.
- Elhadi, A. A., Maarof, M. A., & Osman, A. H. (2012). Malware detection based on hybrid signature behaviour application programming interface call graph. *American Journal of Applied Sciences*, 9(3), 283.
- Roundy, K. A., & Miller, B. P. (2010). Hybrid analysis and control of malware. In *Recent Advances in Intrusion Detection: 13th International Symposium, RAID 2010, Ottawa, Ontario, Canada, September 15-17, 2010. Proceedings 13* (pp. 317-338). Springer Berlin Heidelberg.
- Madan, S., Sofat, S., & Bansal, D. (2022, November). Tools and Techniques for Collection and Analysis of Internet-of-Things malware: A systematic state-of-art review. *Journal of King Saud University - Computer and Information Sciences*, 34(10), 9867–9888. <https://doi.org/10.1016/j.jksuci.2021.12.016>
- Souri, A., & Hosseini, R. (2018, January 12). A state-of-the-art survey of malware detection approaches using data mining techniques. *Human-Centric Computing and Information Sciences*, 8(1). <https://doi.org/10.1186/s13673-018-0125-x>
- M., G., & Sethuraman, S. C. (2023, February). A comprehensive survey on deep learning based malware detection techniques. *Computer Science Review*, 47, 100529.
<https://doi.org/10.1016/j.cosrev.2022.100529>